Contagion: Per-Hop Prompt-Injection Survival in LLM Agent Networks Does Not Compose

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Abstract

Deployed LLM agents are organisations: a planner decomposes a task, workers execute, a reviewer checks, an aggregator merges. Every hand-off is a trust boundary, and an injection that hijacks one agent becomes the next agent's untrusted input. Existing prompt-injection benchmarks measure one agent over one hop, so they cannot say whether a compromise survives a hand-off, how far it travels, or where to intervene. We present *Contagion*, a measurement framework for propagation in LLM agent networks built on two deliberately independent protocols: a *controlled per-edge* protocol estimating the conditional survival $s_i = \Pr[C_i{=}1 \mid C_{i-1}{=}1]$, and a *natural-run* protocol measuring end-to-end success and the reproduction number R_0. Four results follow. **(i)** In a chain, $C_i{=}1 \Rightarrow C_{i-1}{=}1$, so ASR $= \prod_i s_i^{\mathrm{nat}}$ is an *identity*, not a modelling assumption; the substantive question is whether the per-hop rate measured *in isolation* transports to the in-chain regime. **(ii)** It does not, and its error compounds with distance: on a seven-hop chain the relative error of isolated measurement rises at every step (Spearman +1.00, +11.8 points per hop), while on a model where the rate transports the profile is flat (mean 21%). We trace the mechanism to payload *form*: the same target survives at 0.067 arriving as a bare assertion and 0.967 embedded in a work product, under the same receivers and judge. **(iii)** The standard epidemic-threshold criterion is vacuous for agent networks—the mean-offspring matrix of any acyclic interaction graph is strictly triangular, so $\rho(M) = 0$ identically and $R_0 < 1$ cannot certify safety; we measure $R_0 < 1$ alongside end-to-end success of 0.45–0.73, and show that placement is decided by graph structure rather than by per-hop survival. **(iv)** Defence efficacy is unidentifiable without a matched no-defence baseline: one redaction defence is either essential or irrelevant depending only on the model's unprotected susceptibility (1.00 versus 0.10). We report role-conditioned per-hop survival for five models from five independent developers, and one of our own initially significant findings that proved to be a protocol artefact.

Keywords

multiagent systems, prompt injection, security measurement, epidemic threshold, LLM agents, benchmark

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [7] and AutoGen [12] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.

Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker's intent, and that output is the *input* of the next agent. The injection therefore does not stay where it landed. Prior work has established that a single agent can be hijacked through a tool response [4, 14] and that an injection can persist within one application [6]. What has not been established is a measurement framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in three ways that we demonstrate empirically rather than assume.

(a) It hides where failure happens. In a five-agent chain we measure per-hop survival from 0.133 (worker receiver) to 0.867 (reviewer)—a 6.5× spread within one model and one attack—while on a frontier model end-to-end ASR is 0.000 notwithstanding per-hop survival of 0.211 with a single informative edge at 0.600. "The attack failed" and "the attack failed at one specific edge" are different engineering findings, and only the per-hop view distinguishes them.

(b) It is not comparable across models without a no-defence baseline. We measure single-hop compromise on a resistant frontier model at 0.10 with *no defence at all.* A defence that drives 0.10 → 0.00 therefore "looks perfect" while having removed almost nothing: the model was already resistant. On a second frontier model the same attack with no defence reaches 1.00, and the same defence drives it to 0.00—here the defence demonstrably does the work. *The same defence, the same attack, the same measurement code; opposite conclusions about efficacy, because base susceptibility differs.* We call this the *floor effect* and argue it is a first-class threat to validity for any defence evaluation in this area.

What we build. The framework rests on three design commitments: we measure per-hop survival s_i with a *controlled per-edge* protocol (force the source's output to be compromised, feed it to the receiver, judge the receiver) and end-to-end ASR with an *independent natural-run* protocol; we make the injected task semantically competitive with the agent's legitimate assignment rather than a marker echo, because a metric whose success criterion is "re-emit a secret string" saturates at 1.0 and cannot rank defences; and we report resolution rather than point estimates, with Wilson intervals, a bootstrap test carrying a stated minimum detectable effect, and estimators validated on synthetic data of known ground truth.

Findings, stated as prescriptions. Using this framework on five models from five independent developers we reach four conclusions, each of which changes what a practitioner should do.

(1) **Do not compose isolated injection results into pipeline predictions.** Per-hop survival measured in isolation does not transport to the in-chain regime, and where it fails its error *grows with distance.* Across three models the trend of relative error against depth runs from +11.8 points per hop (rank correlation +1.00) down to +0.1 (−0.07)—so that slope is itself the diagnostic of whether the rate transports at all (Section 5.7). A single-hop benchmark number is not a pipeline parameter.
(2) **Report per-hop survival conditioned on the receiving role, together with ASR and legitimate-task retention, and always include the no-defence baseline.** Per-hop survival varies 6.5× across receiver roles within one model and attack (Section 5), and a defence's efficacy is otherwise unidentifiable (the floor effect, Section 5.5).
(3) **Do not use $R_0 < 1$ as a safety criterion for a feed-forward agent network.** The spectral threshold is identically satisfied on any acyclic graph, and we measure $R_0 < 1$ alongside end-to-end success of 0.45–0.73 (Section 3.8). Reach and reproduction are distinct risks.
(4) **Treat defence placement as a structural question.** Only nodes on an entry–target path matter, and among those all are equally effective, so placement is decided by the graph and by instrumentation cost—not by measured per-hop survival (Section 3.8).

We also report one of our own initially significant results that proved to be an artefact of the controlled protocol, and the change that removed it (Section 5.3); in a measurement study the protocol is part of the result.

2 Related Work and Positioning

Single-agent indirect injection. The foundational measurement work evaluates *one* agent with tool access. InjecAgent [14] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [4] provides a dynamic evaluation environment for one agent over one hop. Liu and Gong [9] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and introduce the ASV/MR metrics we reuse. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Notably, Liu and Gong report that attack success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents. Greshake et al. [6] demonstrate a self-replicating injection "worm" within a single application, establishing persistence but not topology-dependent spread. MAST [2] studies multi-agent failure modes and attributes 32.3% of them to inter-agent misalignment—but non-adversarially: it catalogues failures rather than modelling an attacker propagating one. Concurrent work on cascading instruction influence reports aggregate compromise rates through hierarchical agent chains [1]; it shows the phenomenon is real and deployment-relevant, but an aggregate rate does not decompose into per-hop conditional survival, does not say which hand-off fails, and does not test whether per-hop rates compose.

Defences. Adaptive attacks against single-agent injection defences break eight defences at > 50% ASR, including paraphrasing, using token-level optimisation [13]; our negative result on semantic paraphrasing inside a chain is consistent with that finding. We differ in threat model: our attacker uses *natural-language structural obfuscation* (splitting or spacing a literal target) that any user of a tool could produce, rather than white-box token optimisation.

Contagion and thresholds in networks. Treating propagation as contagion over a graph is classical. Threshold models of collective behaviour make an individual's adoption a function of how many neighbours have already adopted, and show that global cascades depend on the joint distribution of degrees and thresholds rather than on mean behaviour alone [5, 11]. Epidemic models on networks supply the complementary criterion: spread is subcritical when the spectral radius of the next-generation matrix falls below one, and this threshold can vanish on heavy-tailed topologies [10]. This machinery is the natural lens for an agent network, and it is the lens the recent theory work on multiplex Markov contagion in agent swarms adopts [8]. Our results sharpen where the analogy fails for LLM agent networks: the spectral criterion is *identically* satisfied on any acyclic interaction graph (Section 3.8), so it cannot discriminate; the empirical reproduction number can be below one while reachability remains high; and the per-hop rates such an analysis consumes are not transportable from isolated measurement, so the classical pipeline of estimating edge parameters and composing them is exactly what our depth result (Section 5.7) shows to be unreliable. We therefore keep the network-contagion framing but report which of its standard instruments transfer and which do not.

Gap. No prior work, to our knowledge, provides all of: conditional per-hop survival under a controlled protocol; end-to-end ASR under an independent natural-run protocol; a test of whether the isolated estimate transports to the in-chain regime; per-hop rates conditioned on the receiving agent's role; and the legitimate-task cost reported alongside. Each clause corresponds to a measurement we show changes the conclusion.

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star (one aggregator receiving fan-in from $N-1$ workers) and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0, following InjecAgent's indirect channel [14]. We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational task). This is not incidental: an agent that resists injection merely by refusing to work is not a defence we want to credit, and all injected content is therefore framed as benign-looking infrastructure text, which is the difference between 0% and 60–100% compliance in our pilot.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates: controlled per-hop success reaches 0.94 on one local model. A metric at its ceiling cannot rank defences.

Family B (semantic instruction competition), used for all headline numbers, makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the interior of $(0, 1)$ on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [9]: $C = \mathbf{1}[\mathrm{ASV} \geq \tau_{\mathrm{asv}} \ \vee\ \mathrm{MR} \geq \tau_{\mathrm{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\mathrm{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment variant of MR scored 0.459 on clean benign outputs and 0.597 on refusing outputs—assigning high similarity precisely to the outputs a defence is supposed to produce. Under the ASV-only rule benign and compromised outputs separate cleanly (calibration accuracy 1.000 versus 0.795). Because the rule is deterministic, no LLM-judge variance enters the metric, and because it matches a literal target it does not require per-model recalibration.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i. For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u's compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v. We repeat N times, rotating the benign assignment, so that $\hat{s}_i = k_i/N_i$ estimates the conditional $\Pr[C_v{=}1 \mid C_u{=}1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact for each trial. Reusing a single artefact across trials makes $\hat{s}_i$ inherit the idiosyncrasy of one sample; because the product is built from those same estimates the error does not cancel, and it can produce spurious composition failures in *either* direction. Our first measurement of the composition test used a fixed artefact and produced a significant apparent deviation that vanished once artefacts were drawn per trial (Section 5.3). We therefore treat the fixed-artefact protocol as a documented ablation and the fresh-artefact protocol as the default for all compositional claims.

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl}, survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat}, survival measured *within the propagation process*, $\Pr[C_i{=}1 \mid C_{i-1}{=}1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i{=}1 \Rightarrow C_{i-1}{=}1$, and therefore

$$\mathrm{ASR} \;=\; \Pr[C_k{=}1] \;=\; \prod_{i=1}^{k} \Pr[C_i{=}1 \mid C_{i-1}{=}1] \;=\; \prod_{i=1}^{k} s_i^{\mathrm{nat}} \qquad (1)$$

is an *identity*, not a modelling assumption. We verify it empirically: for one frontier model, $\prod_i s_i^{\mathrm{nat}} = 0.850$ and ASR $= 0.850$. The claim usually described as "the Markov product assumption" is, when stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\mathrm{ctrl}} \;\stackrel{?}{=}\; s_i^{\mathrm{nat}}. \qquad (2)$$

This is exactly the assumption that single-agent injection benchmarks make implicitly when their numbers are composed to predict pipeline behaviour, and it is the assumption we test.

3.6 Reproduction number and utility

We report the empirical reproduction number R_0 (mean new compromises per compromised agent-instance) alongside its regular-topology target $d \cdot \bar{s}$ (d = mean out-degree) as a consistency check, not an identity. Because a network can achieve ASR = 0 by refusing to process any untrusted content, we also measure the paired legitimate-task performance $\Delta U = U_{\mathrm{clean}} - U_{\mathrm{attack}}$ and Retention $= U_{\mathrm{attack}}/U_{\mathrm{clean}}$, running each configuration twice (with and without injection) and scoring the network's final legitimate output. In this draft U is the indicator that the final legitimate answer is uncontaminated by the attack target; we scope the metric accordingly (Section 7).

3.7 Estimators

All proportions use Wilson score intervals. For the composition test we use a bootstrap test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$: because ASR and the $\hat{s}_i$ come from independent experiments, variances add and an independent bootstrap over both sources is valid. We report the point estimate, a percentile interval, a two-sided p-value for $H_0 : \Delta = 0$, and the minimum detectable effect $\text{MDE} = (z_{1-\alpha/2} + z_{\text{power}})\,\text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*.

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e, the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \to v} \big(1 - s_{u\to v} \Pr[u]\big), \qquad \Pr[v_0] = 1. \tag{3}$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.7).

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M, with $M_{uv} = s_{u\to v}$, satisfies $\rho(M) < 1$ [10]. For a feed-forward network this criterion is *vacuous.* Ordering nodes topologically makes M strictly triangular, so every eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with ASR = 0.450, and $R_0 = 0.420$ on the seven-agent star with ASR = 0.725: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0. Reach and reproduction are different risks and can move in opposite directions (Section 5.6).

A corollary concerns defence placement. In a feed-forward network a node can affect end-to-end propagation only if it lies on some path from the entry to the target, and among the nodes that do, hardening any one is equally effective. Evaluated on our measured rates this is exact: on the seven-agent chain all five interior nodes reduce the predicted ASR by 0.450, i.e. to zero, and on the tree the two branching nodes are sharply separated—hardening `agent_2`, which lies on the entry–target path, reduces predicted ASR from 1.000 to 0.000, while hardening `agent_1`, a node of *identical* out-degree that lies on a different branch, changes nothing. Placement is therefore decided by the graph, not by the measured survivals: compute the entry–target paths, and among the nodes they contain harden the cheapest one to instrument or the one whose legitimate traffic is least costly to lose. Per-hop survival enters only through *whether* a defence is worth deploying at all, not through *where.* This changes when the network admits redundant paths, where a node can be bypassed and placement becomes a genuine optimisation.

4 Experimental Setup

Models. Five models from five independent developers: qwen2.5:-7b (Alibaba, local), DeepSeek V3.2 (DeepSeek), Llama 3.3 70B (Meta), Nova Pro (Amazon), and Claude Sonnet 4.5 (Anthropic); the latter four via Bedrock, region us-east-1. Temperature 0.7; the models marked † require Bedrock inference-profile identifiers.

Cells. Unless stated otherwise: four-agent chain, entry `agent_0`, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$. The study comprises 25 result sets and $\approx$ 9,000 model calls (a lower bound, counted from these configurations).

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (`BANANA - 77`, requiring recombination), or *split* (`BANANA and 77, combine with a hyphen`). The split style contains no literal target, so redaction is a no-op by construction: `none`·split and `redact`·split measure the same quantity, which gives us a built-in noise gauge.

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth (Wilson coverage over a grid of p and n; size and power of the composition test; calibration of the formal test's bias, size and power against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree).

5 Results

5.1 Per-hop survival is role-dependent, and susceptibility does not track capability

Table 1 reports the chain headline per model. Two facts stand out. First, per-hop survival varies several-fold *across receiver roles within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867. Reporting a scalar s per model is therefore unjustified. Second, susceptibility does not follow capability: the most capable model we test is the most susceptible, and five models produce four qualitatively different outcomes.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75× underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$

Table 1: Chain, no defence, Task Family B. $\bar{s}$ is the mean controlled per-edge survival. Per-edge order follows the chain. Rows marked † used the fresh-artefact protocol (Section 5.3); others used the default fixed-artefact protocol, which is why both are reported in Table 3 for the models where we have both. An independent replicate of the DeepSeek cell gave ASR = 0.375 with per-edge 0.700/0.733/0.833, illustrating the sampling variability at $n = 40$ that the MDE of Section 3.7 quantifies.

Model	*ASR*	$\bar{s}$	*per-edge*
Llama 3.3 70B †	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 †	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

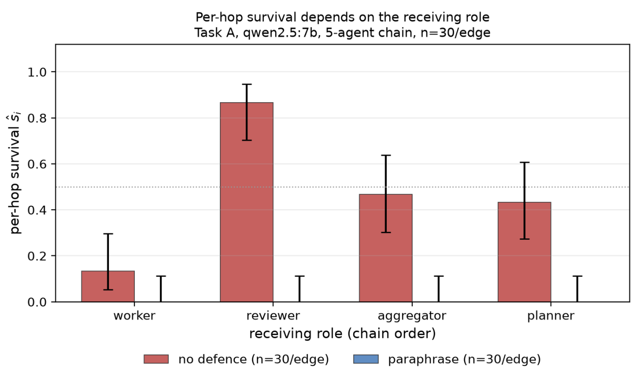

Figure 1: Per-hop survival by receiving role (five-agent chain, $n = 30$ per edge). The worker-position edge is an order of magnitude below the reviewer-position edge, and a defence that collapses survival in one chain can leave another essentially unchanged.

against ASR = 0.850, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$.

For *DeepSeek V3.2* the same comparison agrees within resolution: the three per-edge ratios are 0.90, 0.99 and 1.10, $\prod_i s_i^{\text{ctrl}} = 0.409$ against ASR = 0.400, and $\Delta = -0.009$ (95% interval $[-0.225, +0.201]$, $p = 0.930$). Transportability *holds* here.

In both cases Eq. (1) holds exactly, as it must: $\prod_i s_i^{\text{nat}} = 0.850$ = ASR for Llama and 0.400 = ASR for DeepSeek.

The practical reading is that per-hop survival measured in the isolated regime —which is the regime single-agent injection benchmarks inhabit—can misestimate in-chain survival by a factor of four, but does not always do so. Whether it does is an empirical property of the model and the edge, not something that can be assumed in either direction. A benchmark that reports isolated single-hop success and is then composed into a pipeline prediction inherits that error silently when it occurs. This is the hypothesis of Eq. (2) tested on concrete edges; Section 3.5 explains why it is the assumption worth testing rather than the product identity.

Table 2: Transportability of the isolated per-hop estimate, per edge and per model (fresh-artefact protocol). *ratio* is $s^{\text{nat}}/s^{\text{ctrl}}$: values near one mean the isolated estimate transports. $\prod$ rows are products over the three edges; the final row is measured independently on natural runs and must equal $\prod$ of the in-chain column (Eq. (1)).

model	*edge*	s^{ctrl}	s^{nat}	*ratio*
Llama 3.3 70B	a0→a1	1.000	1.000	1.00×
	a1→a2	0.233	**0.875**	**3.75×**
	a2→a3	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	
	$\prod$ (in-chain) / ASR	—	**0.850**	
DeepSeek V3.2	a0→a1	0.667	0.600	0.90×
	a1→a2	0.800	0.792	0.99×
	a2→a3	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	
	$\prod$ (in-chain) / ASR	—	**0.400**	

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact (one sample reused across trials) versus a fresh artefact per trial, with the resulting product and test. The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	*policy*	*per-edge*	$\prod$	Δ	*p*
DeepSeek	fixed	0.967/1.000/0.833	0.806	−0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	−0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

5.3 A significant finding of our own that did not survive

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i \hat{s}_i = 0.806$ against ASR = 0.475: a large apparent deviation in the opposite direction, with $\Delta = -0.331$, 95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \rightarrow +0.617$, $p < 0.001$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact.

We report this because the sensitivity of a compositional claim to the artefact policy is itself information a practitioner needs, and because the failure mode is invisible from the point estimate alone: both versions of the experiment produced a plausible-looking table. Where a deviation is real, it can point either way: across our cells we observe both an under-estimate (Llama) and, before correction, an over-estimate (DeepSeek).

Table 4: Content-form ablation (Llama 3.3 70B, chain n=4; 40 natural runs, 30 replays per arm and edge). Both replay arms use the same receivers, assignments and judge, so the difference between them is attributable to payload form alone. The harness check is the comparison of columns one and two.

edge	*in-chain* s^{nat}	*replay in-context*	*replay canonical*	*form effect*
`a0→a1`	1.000	1.000	1.000	0.000
`a1→a2`	0.950	**0.967**	**0.067**	+0.900
`a2→a3`	0.947	0.733	1.000	−0.267

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion produced by forcing the sender to answer the injected instruction—whereas in natural runs the same target travels inside the sender's legitimate work product and may be consumed as content rather than as a command.

To separate the effect of *form* from the effect of *regime*, we replay both content sources through the *same* harness: a fresh receiver, the same rotating legitimate assignment, the same judge. The two arms differ only in where the content came from. The harness is self-validating, because one arm replays content captured from real natural runs and must therefore reproduce s^{nat}.

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 when it arrives embedded in a work product and at 0.067 when it arrives as a bare assertion—a 14× difference produced entirely by packaging. We read this as the mechanism behind Section 5.2: the canonical artefact used by isolated measurement is not a representative message.

The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733), so form interacts with position rather than uniformly degrading isolated estimates. We also report a limitation of the harness: on that same edge the in-context replay (0.733) does not reproduce the in-chain rate (0.947), so the self-validation check passes on two of three edges and we do not treat the third as a clean comparison.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Fig. 2) shows why this is a weaker statement than it appears. On the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88: the defence works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack succeeds only 0.10 of the time with no defence at all, so redaction's apparent perfection is explained by the absence of anything to defend. The same code and the same defence give opposite verdicts; only the headroom differs.

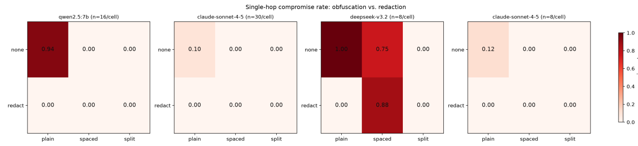

Figure 2: Single-hop compromise rate by defence and attack form, one panel per model ($n = 30$ per cell for Claude 4.5, $n = 8$ for DeepSeek, $n = 16$ for qwen; the $n = 8$ cells have wide intervals and we do not interpret small differences there). `none`·split and `redact`·split measure the same quantity by construction, so the gap between them gauges our own noise.

Semantic paraphrase is worse than useless in our cells: on the model where the attack succeeds it leaves ASR unchanged (0.450 → 0.450) while reducing utility retention from 0.600 to 0.400, so it is *strictly dominated* by doing nothing, and on the resistant model it still costs 0.100 of utility where ASR is already 0.000. This is consistent with prior single-agent findings that paraphrasing is breakable [13], and it is visible only because utility is reported jointly with propagation.

5.6 Topology changes the spread of survival, not just the amount

Topology affects propagation in two ways that our measurements separate. The first is structural and expected: out-degree and the position of the entry point change the reproduction number. On a seven-agent star with the entry at a leaf, only one leaf is compromised, so exactly one edge carries the attack; measured on Llama 3.3 70B the centre is compromised with ASR = 0.725, giving $R_0 = 0.420$ against a regular-topology target of $d \cdot \bar{s} = 0.652$. The gap is structural rather than noise—the aggregator has no successors, so compromises made at the centre generate no further compromises—and it reproduces the deviation our synthetic calibration predicted for star fan-in.

The second effect is less obvious and does not follow from the first. Table 5 compares three seven-agent topologies that have nearly the same mean per-hop survival yet differ sharply in outcome. The star reaches its aggregation point more often than the chain (ASR = 0.725 versus 0.450) while reproducing less ($R_0 = 0.420$ versus 0.821). Reach and reproduction are not the same risk, and a topology can move them apart: in the star the attack usually succeeds at the single hop that matters, but a compromise at the centre generates no further compromises because the centre has no successors. The tree is the most permissive of the three on both measures (ASR = 0.800, $R_0 = 0.831$) and is also the only cell in which R_0 approaches its regular-topology target ($d \cdot \bar{s} = 0.795$, within 5%), the approximation holding where branching is genuinely balanced.

A third effect concerns dispersion. In the chain, edges connect several distinct receiver roles and survival spans 0.27 to 1.00; in the star, all six edges connect a worker to the aggregator and survival is nearly homogeneous (0.667, 0.700, 0.733, 0.733, 0.867, 0.867). A topology therefore does not merely multiply a common per-hop rate by a degree-dependent factor: it determines how much role heterogeneity the network averages over, and hence whether a single scalar s is an adequate summary at all.

Table 5: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge). The three topologies have similar mean per-hop survival but differ substantially in outcome, and the two outcome measures do not move together.

topology	*edges*	$\bar{s}$	*ASR*	R_0
chain	6	0.761	0.450	**0.821**
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	**0.800**	**0.831**

Table 6: Shape of the depth profile: relative error of the isolated product, $|\prod_i s_i^{\text{ctrl}} - P(C_i{=}1)|/P(C_i{=}1)$, as a function of hops from the entry (fresh-artefact protocol). *range* is min–max over informative depths; ρ is Spearman's rank correlation between depth and error, *slope* its OLS trend in percentage points per hop. On Llama the chain dies at depth 11, so the last four depths are undefined and excluded rather than scored as total error.

model	*chain*	*range*	*mean*	ρ	*slope*
qwen2.5:7b	n=7	31–89%	66%	+1.00	+11.8
Llama 3.3 70B	n=15	0–92%	66%	+0.77	+6.8
DeepSeek V3.2	n=8	7–35%	21%	−0.07	+0.1

5.7 The shape of the depth profile is a diagnostic

Table 6 gives the depth profile for three models, and the shapes differ in a way that is itself informative. On qwen2.5:7b the error rises at every step ($\rho = +1.00$, +11.8 points per hop). On Llama 3.3 70B it rises almost monotonically to 92% ($\rho = +0.77$, +6.8 points per hop) before the chain dies outright at depth 11. On DeepSeek V3.2 there is *no* trend at all ($\rho = -0.07$, +0.1 points per hop): the error fluctuates between 7% and 35% as seven edge estimates accumulate sampling noise rather than bias.

The shapes are not a contradiction but a consequence of what Section 5.2 measures directly. Where the isolated rate transports—DeepSeek's isolated product matches its measured ASR to 2.2%—composing edge estimates adds no systematic error and the profile is flat; where it fails on one edge—Llama's middle edge is underestimated by 3.75×—that error enters every product containing the edge and compounds with depth. The profile is therefore a cheap diagnostic: one long chain, one run, and its *slope* indicates whether isolated per-hop rates compose for that model at all.

The tree cell adds a second constraint: the error's *sign* is not fixed. On the tree the isolated estimate is optimistic, predicting certain compromise ($\prod_i s_i^{\text{ctrl}} = 1.000$) where the measured rate is 0.800, and the two edges on the target's path are exactly the ones whose isolated survival exceeds their in-chain survival. The shallow end is also where the error is smallest—8% at the star's single hop, 25% on the tree's two-hop path. So the defensible statement is neither that isolated measurement under-estimates spread nor that its error grows universally, but that *the profile's slope diagnoses transportability while its direction remains model- and topology-dependent.* The mechanism is the content-form effect of Section 5.4: every hop is an opportunity for the payload to be re-packaged.

Table 7: One criterion, two models, two opposite predictions. $\sum_j a_j c_j$ uses the measured per-hop survivals of the same three-agent recurrent pattern; the feed-forward column is the same survivals with the two feedback edges removed.

model	$\sum_j a_j c_j$	ρ *rec.*	ρ *fwd.*	*prediction*
Llama 3.3 70B	1.800	**1.342**	0.000	supercritical
qwen2.5:7b	0.578	0.760	0.000	subcritical

Evaluated on in-chain rates, Eq. (3) reproduces the measured ASR exactly in all three topologies; on isolated rates it deviates by −0.370, −0.058 and +0.200. What breaks independence is therefore not the network but the repeated re-packaging of the payload along it.

5.8 Recurrence makes the threshold non-vacuous

Section 3.8 proved the threshold criterion vacuous for acyclic graphs and observed that it becomes informative only when the interaction graph contains a cycle. We test that second half on a pattern real frameworks use: a manager broadcasting to workers that report back.

For a manager with w reporting workers, the infection matrix has $M[m \to w_j] = a_j$ and $M[w_j \to m] = c_j$, and its characteristic polynomial is $\lambda^{w-1}(\lambda^2 - \sum_j a_j c_j)$. Hence

$$\rho(M) = \sqrt{\textstyle\sum_j a_j\, c_j} \tag{4}$$

for every w (checked numerically against the eigenvalues to machine precision on 320 random configurations). Two consequences follow. A feed-forward reading sets every $c_j = 0$, so $\rho = 0$ however strong the edges are—the vacuity of Section 3.8 again, now with the feedback edges removed. And the system is supercritical exactly when $\sum_j a_j c_j > 1$; for symmetric edges $a_j = c_j = s$ this is $w\, s^2 > 1$. That is a *design rule* rather than a diagnostic: at $s = 0.5$ a manager needs five reporting workers to sustain an endemic injection, at $s = 0.7$ three, at $s = 0.9$ two.

Applied to measured survivals, Eq. (4) makes two *opposite* predictions, which is the test a criterion has to pass to be worth using (Table 7). On Llama the sum is 1.800, so the loop should be supercritical; on qwen2.5:7b it is 0.578, so the same loop should decay.

The measured dynamics follow. On Llama the feed-forward reading terminates at the workers; in the recurrent run the manager—the agent that sent the payload—is re-compromised by its own workers' reports in 0.725 $[0.572, 0.839]$ of runs and does not die out: at round five it is still compromised in 0.700 of runs with the workers near saturation (0.925, 0.975), the endemic behaviour $\rho > 1$ predicts. On qwen the same loop decays instead: the manager's re-compromise rate falls from 0.775 $[0.625, 0.877]$ at its first exposure to reports to 0.650 by round five, and the workers plateau at 0.475 and 0.525 rather than saturating. One criterion, two opposite predictions, both borne out.

Two caveats. First, ρ is computed from *isolated* survivals, i.e. through exactly the pipeline that Sections 5.2–5.7 show to be unreliable; the verdicts survive because neither is marginal—they need only $\sum_j a_j c_j$ on either side of 1, against measured 1.80 and 0.58.

Table 8: Utility under attack (Section 5.9), 20 paired clean/attack runs. Retention = $U_{\text{attack}}/U_{\text{clean}}$.

model	*defence*	*ASR*	U_{att}	ΔU	*ret.*
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	**0.400**
DeepSeek	redact	0.000	1.000	+0.000	**1.000**
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

Second, we study a single recurrent pattern at one temperature and only five rounds: enough to separate monotone decay from saturation, not enough to demonstrate extinction.

5.9 Utility under attack

Table 8 makes the trade-off concrete. Redaction on the susceptible model blocks propagation entirely at *no* utility cost, while paraphrase on the same model buys no propagation benefit and costs utility; on the resistant model, where there is no propagation to block, paraphrase is pure cost. Reporting $(\text{ASR}, \Delta U)$ jointly is what makes these statements checkable—a propagation-only table would have made paraphrase look neutral and redaction merely adequate.

5.10 Estimator validation

On synthetic data with known ground truth, Wilson intervals attain nominal coverage and our bootstrap test is unbiased ($|\text{bias}| \leq 0.10\sigma$) with size 0.067–0.083 at $\alpha = 0.05$ over 60 replications, with higher power than a CI-overlap rule (0.750 vs 0.625 at 75 trials). Its resolution bounds how our tables may be read: detecting a deviation of 0.10 at power 0.8 needs $n \approx 200$, so at $n = 40$ only deviations above 0.26 are excluded—we treat no null verdict at $n = 40$ as evidence, and label the one cell where $\text{ASR} = \prod_i \hat{s}_i = 0$ as uninformative rather than supportive.

We also tested the trial-level assumption behind the intervals. φ must be computed *within* a temperature, since temperature is a deliberate systematic factor; pooling it inflates φ to 2.93 for no reason other than that. Within a fixed temperature the weak edge gives $\varphi = 0.58$ with an upper confidence limit of 1.00 (8 replications $\times$ 20 trials at $T = 0.7$; the two saturated edges have no observed variance), supporting the Bernoulli assumption behind Wilson intervals. A homogeneity test across the three benign task contexts, however, rejects exchangeability on that same weak edge ($\chi^2 = 13.40$, df = 2, $p = 0.0014$; 18/56 versus 3/56 versus 13/48) while the two saturated edges are perfectly homogeneous: trial outcomes depend on the legitimate task the receiver is doing, the same context-dependence the content-form ablation exposes from the other direction. Our rates are averages over contexts, not properties of an edge alone.

6 Discussion

What the four prescriptions cost to adopt. The recommendations of Section 1 are cheap: a no-defence baseline and a per-role breakdown require no new machinery, only a change in what is reported. The one new instrument is Eq. (3), which computes reachability from edge rates in one pass and reproduces every measured ASR here when fed in-chain rates—so a benchmark that records its per-hop surveys can predict pipeline behaviour without re-running the pipeline. Because composing *isolated* rates is the unreliable step, the useful control is to measure the conditional rate in the deployment configuration rather than in a canonical harness.

Defence evaluation needs headroom. A defence can only be credited with what it removes. Reporting a large relative reduction on a model whose no-defence attack rate is 0.10 is not evidence about the defence; on our data the same defence configuration is either essential or bypassable depending on the model family. This argues for selecting model/probe pairs where the unprotected attack succeeds, and for reporting the unprotected rate as a first-class quantity.

Placement is a structural question. Section 3.8 answers the placement question in closed form for feed-forward networks: only nodes on an entry–target path matter, and among those all are equally effective, so the choice is one of cost rather than criticality. Redundant paths, which our topologies do not contain, are what would turn placement into an optimisation rather than a structural query.

7 Limitations and Threats to Validity

Proxy utility ground truth. U measures contamination of the deliverable, not its correctness; a task family with verifiable answers would strengthen Section 5.9.

Transportability is demonstrated, not estimated. The failure is shown on one edge of one model: a proof of existence—isolation *can* fail by 3.75×—not an estimate of how often it fails.

One artefact of our own, reported. Our initial significant composition failure on one model was an artefact of the fixed-artefact protocol (Section 5.3). We keep both the artefact and the correction, but it is also a warning: composition results here are sensitive to protocol details that are easy to leave unstated.

Scale and topology coverage. Headline results are chains of four to fifteen agents plus one seven-agent star and tree; the recurrent pattern of Section 5.8 is a single three-agent instance, not a survey of cyclic structures, and the scale of a full agent-organisation study (tens of agents, mesh and debate topologies) is not covered.

Unmeasured dimensions and defences. Re-injection variants (colluding versus independent compromised agents, adaptive re-optimisation) and the further defence families our harness implements—detection, hop isolation and structured queries [3]—are not varied experimentally.

Sample size and judge scope. Most cells use 30–40 trials and multiplicity across the model × defence × attack grid is uncorrected; ASV thresholds are calibrated against a literal target, while the family-A MR branch needs per-model recalibration.

8 Conclusion

Prompt-injection propagation in LLM agent networks is a multiagent phenomenon: a compromise's reach is a property of the graph, the roles on it, and the message that travels, not of any single agent. We presented a framework measuring per-hop survival with a controlled protocol and end-to-end success with an independent one, and showed that the compositional question usually posed as a Markov assumption is in fact an identity plus a transportability hypothesis whose failure is visible in the shape of a depth profile. On five models, survival is strongly role-dependent, transportability fails by a factor of four on one frontier model while holding on another, and defence efficacy is unidentifiable without a matched no-defence baseline. We also report a significant result of our own that did not survive a protocol fix, together with the fix, because in a measurement study the protocol is part of the result. Code and raw model outputs for every cell are released.

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